

NOVA

IMS

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School

MDSAA

Master's Degree Program in
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Deep Learning

Classifying Animals using Computer Vision

Paulo Martins, r2015469

Beatriz Fonseca, r20201599

Eldar Medvedev, r20181162

Group 69

NOVA Information Management School
Instituto Superior de Estatística e Gestão de Informação

Universidade Nova de Lisboa

Project available in [GitHub](#)

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1. INTRODUCTION

Using images of rare species from the Encyclopedia of Life (Stevens et al. (2024)), the team attempted to predict the animals' families by training a computer vision model. The best model is a pre-trained model on the ImageNet Dataset, by means of EfficientNetB4. To this pre-trained model, the team added Global Average and Max Pooling layers, a Dense layer with 64 neurons and relu activation function, a Batch Normalization layer, another Dense layer with 64 neurons and relu activation function and a final output Dense layer with as many neurons as the family classes and a softmax activation function. This model was trained on preprocessed and upsampled images, in an attempt to correct any noise in the images and fix the major class imbalance of the dataset, respectively.

The results demonstrate that training Convolutional Neural Networks (CNN) from scratch for image classification tasks is very challenging as the team failed to obtain any meaningful accuracy (above 50%), whereas in the case of pre-trained models, the most significant steps were image preprocessing and standardization, despite that larger models, such as EfficientNetB4 vis. EfficientNetB0 will always show better results on the same data.

Regarding the fine-tuning of pre-trained models, it was found that, given the depth of the original model, the number of samples required to improve recall is significant, as false positives, in the presence of many classes, become increasingly likely; thus, for the same reason, regularization and Dropout layers proved ineffective, as the models struggled with hard examples without what was, in the team's understanding, enough data.

2. EXPERIMENTAL SETUP

The team used *Keras* to develop the models and resize the images, and the *cv2* library for the image preprocessing and upsampling. 'pandas' was used to read the metadata files and feed them into the **flow_from_dataframe** *Keras* function. *sklearn* was used to compute some metrics and *matplotlib* to plot a visual representation of model metrics. The *nbformat* library was used to run notebooks inside others.

In addition to these well-known libraries, the team created their own library – *deep* - with all the necessary modules and scripts to reproduce the experiment, which can be installed by running **"pip install ."** in the terminal of the root folder.

To start reproducing the project, the user first needs to install the necessary libraries and then run the command **"setup"** on the terminal to get the project structure ready. When needing to reset the images folders to run any model, the user just must run on the terminal **"reset input"** to clean the **input_image_directory** folder and/or **"reset processed"** to clean the **processed_image_directory** folder.

3. APPROACH EVALUATION

As the project kicked off, the team started by testing some Minimum Viable Product (MVP) models with the raw images using the VGG-16 pre-trained model. This model yielded poor results, with accuracy below the 10% threshold, and, therefore, was quickly discarded and the decision was made to re-evaluate the approach.

As per Sadeghi et al. (2021), and a consensus when browsing the internet, the team realized the data could benefit from preprocessing. Moreover, given the long-tailed distribution of the labels, as is defined in Cunha et al. (2023), some steps towards class imbalance needed to be taken, including album augmentation techniques, as is suggested in Islam et al. (2023).

Inspired by the popular handbook *Machine Learning Yearning* (Ng, A., 2018) and, after thorough analysis of the dataset, as Islam et al. (2023) also suggests, the team hypothesized that training a model specific to the task of recognizing the presence of animals in a picture could be advantageous and assist a main model in identify animal family, since the dataset contained several examples of non-animal images, e.g. animals skulls and skeletons, and of journals; such that correctly identifying those would yield better results.

Image preprocessing included a cleaning step with the application of median blur, gaussian blur and Contrast Limited Adaptive Histogram Equalization (CLAHE) and a standardization step that made use of ImageNet's mean and standard deviation. Finally, the images were resized to the select pre-trained model's specified size¹ and the pixel values were normalized to values between 0 and 1.

3.1. ANIMAL VS. NO ANIMAL MODEL

To develop the animal vs. no animal model, the team started by developing a tool to classify the images on this new label, the 'label_binary.py'. Then, the EfficientNetB4 pre-trained model was used with some extra layers, namely a Global Average and Max Average Pooling layers, a simple Dense layer with 128 neurons and a relu activation function, a Batch Normalization layer, a Dense layer with 64 neurons, and a final Dense layer for binary output with 1 neuron and a sigmoid activation function. The model was tested with different loss functions - binary cross entropy and focal loss (as in Lin et al. (2017)) -, kernel l2 regularization, and a Dropout layer.

This model was tested with the raw images, the preprocessed images and preprocessed upsampled images. To upsample the images, the team took two approaches - 1) the use of hard negatives cropped from the background of images where animals were known to be present², and 2) via the use of augmentation techniques to oversample the data both on- and offline³.

3.2. FAMILY MODEL

Similarly to the binary model's case described above, models were iteratively tested using combinations of pre-trained models with frozen layers and a varying number of Dense layers; these

¹ These configurations were made changeable using the development framework with the *deep* package, and meant that, for any particular model to be trained, the images were resized accordingly.

² See 'label_cropper.py', a tool for background cropping.

³ See 'album_augmenter.py', a tool for offline image augmentation.

Dense layers themselves, of varying size. Beyond variations in width and depth, L2 regularization, early stopping, learning rate decay and the addition of a Dropout layer were all attempted.

Regarding pre-trained models, as the VGG family had already proved unsatisfactory, an initial implementation with EfficientNetB0 quickly convinced the team to make use of EfficientNetB4, since it was found to be a good compromise between the team's time and hardware constraints and the pursuit of quality in the results.

The images were preprocessed and upsampled, using the same scaffolding as the binary model's, yielding different results.

As was mandated, the team developed a CNN model from scratch, following loosely the VGG-16 architecture. The effort was placed on understanding the role of the layers and how they contribute to the final result, more so than actually training the model for quality results, since a quick inspection of any pre-trained model will show a rather deep architecture, which was infeasible to construct using the available processing resources and impractical to train given the data availability.

4. RESULTS EVALUATION

4.1. ANIMAL VS. NO ANIMAL MODEL

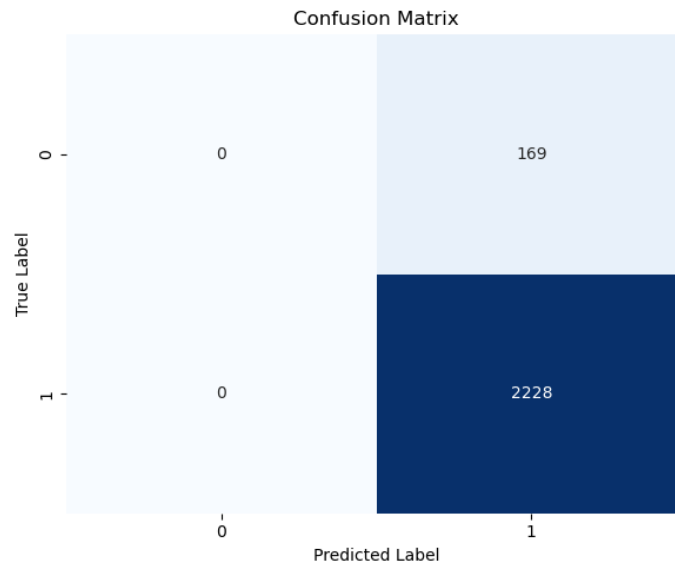
The use L2 kernel regularization and a Dropout layer did not show any model improvement. Even though focal loss yielded better loss values, the team valued the Area Under Curve values more and decided to go with the binary cross entropy.

The best performing model on the validation set was the one that used the upsampled images; however, it proved to have a lot of underfit between the train and validation sets and, therefore, the model with only the preprocessed images was selected to make the final predictions on the test set.

This model got an AUC of 0.621 during training, which proves it can predict a bit better than a random model but is still very uncertain of the decisions (since this value is below the 0.7 threshold). Below is a table with the metrics from the train and test sets:

<i>Dataset</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>
<i>Train</i>	0.9294	0.9293	1
<i>Test</i>	0.9295	0.9295	1

The results seemed overall satisfying, but all hope was lost when the confusion matrix was analyzed – the model was unable to learn the 'no animal' label, predicting all images as 'animal'. Therefore, the team forgone the idea of using a composite model with this label, focusing solely on the family task.



4.2. FAMILY MODEL

Animal recognition, as a subset of computer vision, often exhibits long-tailed distributions of class labels, as discussed in Cunha et al. (2023). This was precisely the case in our project. As highlighted by Wang et al. (2022), the core challenge in long-tailed learning lies in a model's ability to establish robust decision boundaries in scenarios where many classes are underrepresented and often visually similar. In our case, this issue manifested clearly - initial results were weak, even using our best-performing models.

The introduction of multiclass focal loss was pivotal. Prior to its implementation, our models exhibited high precision but struggled with recall, learning readily from dominant, easy examples but failing to generalize well to rarer classes. Focal loss helped rebalance this by encouraging the model to focus on harder, misclassified examples.

These dynamics illustrate both the complexity of long-tailed decision boundaries and a promising direction for addressing them. Our findings reinforce focal loss not just as a workaround but as a crucial tool in equitable model training for imbalanced vision tasks.

Thus, the best approach implemented multi-class focal loss in conjunction with the aforementioned preprocessing schema, with identical model specification to the binary case, changing only the output layer for one with as many neurons as the number of classes and a softmax activation function. The team makes note that no model specification including L2 regularization and a Dropout layer with rate above 0.1 ever showed signs of convergence.

The relevant statistics for this model can be seen in the table below:

<i>Dataset</i>	<i>Train</i>	<i>Validation</i>	<i>Test</i>
<i>Accuracy</i>	0.7903	0.6505	0.6725

Even though this model shows a bit of overfit on the train data, it was the one that yielded the best results – due to the familiarity with the data, the team is certain the overfit could have been mitigated by implementing an early stop callback on the model.

Below are the model’s top 5 mispredicted classes and their respective True Positive and False Negative rates. The full graph can be consulted in the codebase.



5. FUTURE WORK

The results obtained and the possibilities provided by the project’s codebase, had the team raise more questions than answers, from which several possibilities highlighted in the references arise.

In terms of implementation quality, perhaps for such a refined approach, *PyTorch* could have been more suitable. In terms of solution quality, more rigorous and strenuous testing of all image treatment hypothesis would be welcomed, as well as the implementation of hyperparameter tuning via *Keras Tuner* or analogous in an alternate library or framework, since it could lead to a more optimized solution in terms of model finetuning – i.e., regarding layers’ width, depth and type, among other hyperparameters.

Moreover, after using hand-made python scripts to label dataset images regarding animal presence, the opportunity to use tools such as CVAT and FiftyOne, could prove extremely valuable in assisting with this task, making annotation both richer and more effective, enabling the use of other algorithms, such as RCNN for feature extraction.

Lastly, as is highlighted in Cunha et al. (2023), a back-bone model with several specialized branches could yield better results given the long-tailed distribution of the data labels and, perhaps, even enable a truly multi-label image classification model.

All these hypotheses seem equally interesting and would warrant the team’s full attention and commitment, should the opportunity to work on them arise.

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